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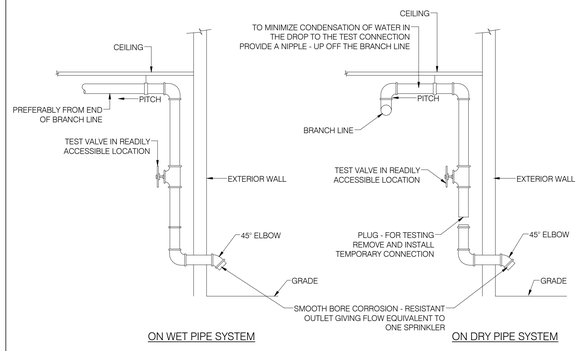
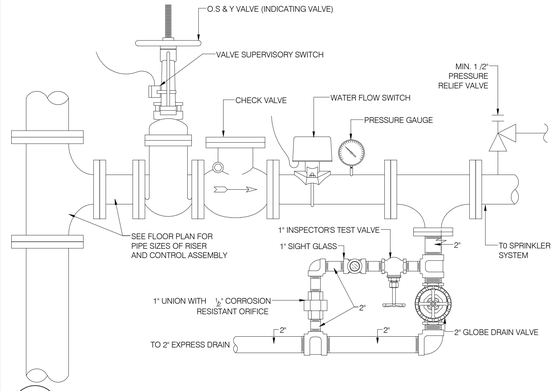


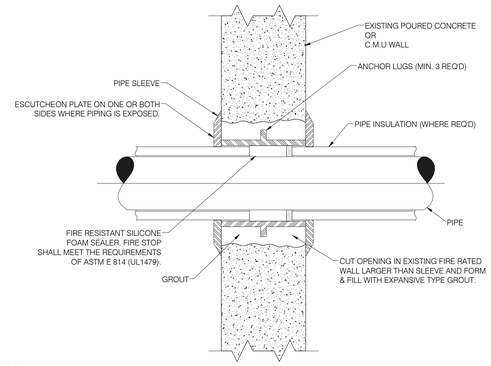
3 FP-7.1 INSPECTORS TEST CONNECTION DETAIL

SCALE: NOT TO SCALE



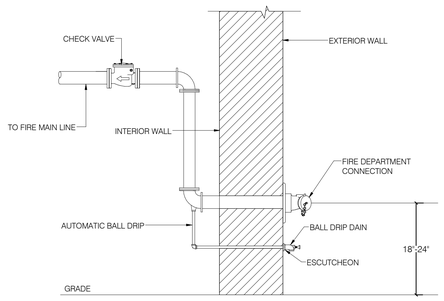
2 FP-7.1 SPRINKLER ZONE CONTROL WITH TEST/DRAIN ASSEMBLY

SCALE: NOT TO SCALE



1 FP-7.1 PIPE PENETRATION THROUGH FIRE RATED WALL

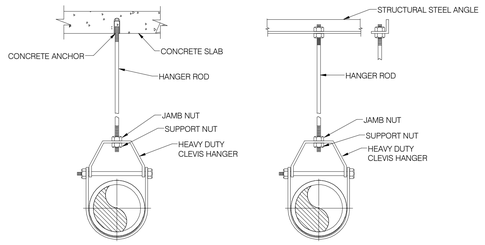
SCALE: NOT TO SCALE



WALL MOUNT

6 FP-7.1 WALL MOUNTED FIRE DEPARTMENT CONNECTION DETAIL

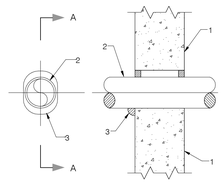
SCALE: NOT TO SCALE



MAXIMUM DISTANCE BETWEEN HANGERS (FEET - INCHES)												
NOMINAL PIPE SIZE (IN INCHES)												
	3/4	1	1 1/4	1 1/2	2	2 1/2	3	3 1/2	4	5	6	8
STEEL PIPE EXCEPT THREADED LIGHT WALL	N/A	12 - 0	12 - 0	15 - 0	15 - 0	15 - 0	15 - 0	15 - 0	15 - 0	15 - 0	15 - 0	15 - 0
THREADED LIGHTWALL STEEL PIPE	N/A	12 - 0	12 - 0	12 - 0	12 - 0	12 - 0	12 - 0	N/A	N/A	N/A	N/A	N/A
CPVC	5 - 6	6 - 0	6 - 6	7 - 0	8 - 0	9 - 0	10 - 0	N/A	N/A	N/A	N/A	N/A
COPPER TUBE	8 - 0	8 - 0	10 - 0	10 - 0	12 - 0	12 - 0	12 - 0	15 - 0	15 - 0	15 - 0	15 - 0	15 - 0

5 FP-7.1 PIPE SUPPORT DETAIL

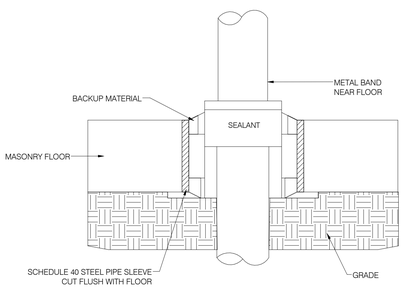
SCALE: NOT TO SCALE



SECTION A-A

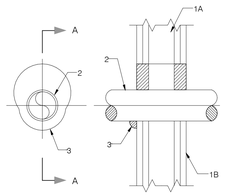
NOTE:

1. WALL ASSEMBLY- MIN 5" (127mm) THICK REINFORCED LIGHT WEIGHT OR NORMAL WEIGHT (100-150 PCF OR 1600-2400 KG/M3) CONCRETE. WALL MAY ALSO BE CONSTRUCTED OF ANY UL CLASSIFIED BLOCKS, MAX. DIA. OF OPENING IS 3" (76mm). SEE CONCRETE BLOCKS (CAZT) CATEGORY IN THE FIRE RESISTANCE DIRECTORY FOR NAMES OF MANUFACTURERS.
2. THROUGH PENETRANTS - ONE NONMETALLIC PIPE TO BE INSTALLED EITHER CONCENTRICALLY ECCENTRICALLY WITHIN THE FIRST FIRE STOP SYSTEM. THE ANNULAR SPACE BETWEEN PIPE AND THE PERIPHERY OF THE OPENING SHALL BE MIN. 0" (POINT CONTACT) TO A MAX. 5/8" (16mm). THE FOLLOWING TYPESS AND SIZES OF THE NONMETALLIC PIPES MAY BE
   USED: A. POLYVINYL CHLORIDE (PVC) PIPE - NOM. 2" (51mm) DIA. (OR SMALLER) CELLAR OR SOLID CORE SCHEDULE 40 (OR HEAVIER) PIPE FOR USED IN CLOSED (PROCESS OR SUPPLY) OR VENTED (DRAIN, WASTE OR VENT) PIPING SYSTEMS.
   B. CHLORINATED POLYVINYL CHLORIDE (CPVC) PIPE - NOM. 2" (51mm) DIA. (OR SMALLER) SDR 135 CPVC PIPE FOR USED IN CLOSED (PROCESS OR SUPPLY) PIPING SYSTEMS.
   C. CROSSLINKED POLYETHYLENE (PEX) TUBING - NOM. 2" (51mm) DIA. (OR SMALLER) SDR 9 PEX TUBING FOR USED IN CLOSED (PROCESS OR SUPPLY) PIPING SYSTEMS.
   D. RIGID NONMETALLIC CONDUIT (RNC) - NOM. 2" (51mm) DIA. (OR SMALLER) SCHEDULE 40 PVC CONDUIT INSTALLED IN ACCORDANCE WITH ARTICLE 347 OF NATIONAL ELECTRICAL CODE.
3. FILL, VOID OR CAVITY MATERIAL* - SEALANT - MIN 5/8" (16mm) THICKNESS OF FILL MATERIAL APPLIED WITHIN THE ANNULUS, FLUSH WITH BOTH SURFACES OF WALL AT POINT CONTACT LOCATION. A MIN 1/2" (13mm) DIA. BEAD OF FILL MATERIAL SHALL APPLIED TO THE WALL/PENETRANT INTERFACE ON BOTH SURFACES OF THE WALL.



7 FP-7.1 PIPE PENETRATION THROUGH FLOOR

SCALE: NOT TO SCALE



SECTION A-A

NOTE:

1. WALL ASSEMBLY- THE 1 AND 2 HR FIRE-RATED GYPSUM BOARD/STUD WALL ASSEMBLY SHALL BE CONSTRUCTED OF THE MATERIALS AND IN THE MANNER DESCRIBED IN THE INDIVIDUAL U300 OR U400 SERIES WALL AND PARTITION DESIGNS IN THE UL FIRE RESISTANCE DIRECTORY AND SHALL INCLUDE THE FOLLOWING CONSTRUCTION FEATURES:
   A. STUDS - WALL FRAMING SHALL CONSIST OF EITHER WOOD STUDS OR STEEL CHANNEL STUDS. WOOD STUDS TO CONSIST OF NOM. 2"x4" LUMBER SPACED 16" OC. STEEL STUDS TO BE MIN 2-1/2" WIDE BY 1-3/8" DEEP CHANNELS SPACED MAX 24" OC.
   B. GYPSUM BOARD* - THE GYPSUM WALLBOARD TYPE, THICKNESS, NUMBER OF LAYERS AND FASTENERS TYPE AND SHEET ORIENTATION SHALL BE AS SPECIFIED IN THE INDIVIDUAL U300 OR U400 SERIES DESIGN IN THE UL FIRE RESISTANCE DIRECTORY. MAX. DIA. OF OPENING IS 3".
   THE HOURLY 'F' AND 'T' RATING OF THE FIRE STOP SYSTEM IS EQUAL TO THE HOURLY FIRE RATING OF THE WALL ASSEMBLY IN WHICH IT IS INSTALLED.
2. THROUGH PENETRANT - ONE NONMETALLIC PIPE, CONDUIT OR TUBING INSTALLED EITHER CONCENTRICALLY OR ECCENTRICALLY WITHIN THE FIRESTOP SYSTEM. THE ANNULAR SPACE BETWEEN PIPE AND THE PERIPHERY OF THE OPENING SHALL BE MIN. 0" (POINT CONTACT) TO A MAX. 1-1/4". PIPE TO BE RIGIDLY SUPPORTED ON BOTH SIDES OF WALL ASSEMBLY. THE FOLLOWING TYPES AND SIZES OF NONMETALLIC PIPES MAY BE USED:
   A. CHLORINATED POLYVINYL CHLORIDE (CPVC) PIPE - NOM. 2" (51mm) DIA. (OR SMALLER) FLOWGUARD GOLD SDR11 CPVC PIPE FOR USED IN CLOSED (PROCESS OR SUPPLY) PIPING SYSTEMS.
   B. CHLORINATED POLYVINYL CHLORIDE (CPVC) PIPE - NOM. 2" (51mm) DIA. (OR SMALLER) BLAZEMASTER SDR13.5 CPVC PIPE FOR USED IN CLOSED (PROCESS OR SUPPLY) PIPING SYSTEMS.
3. FILL, VOID OR CAVITY MATERIAL* - SEALANT - MIN 5/8" AND 1-1/4" THICKNESS OF FILL MATERIAL APPLIED WITHIN THE ANNULUS, FLUSH WITH BOTH SURFACES OF WALL FOR 1 AND 2 HR RATED ASSEMBLIES RESPECTIVELY, AT A POINT CONTACT LOCATION, A MIN. 1/2" DIA. BEAD OF FILL MATERIAL SHALL BE APPLIED TO THE WALL/PENETRANT INTERFACE ON BOTH SURFACES OF THE WALL.

4 FP-7.1 PIPE PENETRATION THROUGH FIRE RATED WALLS

SCALE: NOT TO SCALE

SETTY
ONE SOUTH STREET, SUITE 1130
BALTIMORE, MARYLAND 21202
TEL 667-329-5035
WATER FOUNTAIN SANITARY SYSTEMS MANAGEMENT
ENERGY MANAGEMENT SERVICES • PROGRAM MANAGEMENT
CITY OF BALTIMORE • BALTIMORE COUNTY • PRINCE GEORGES COUNTY
DISTRICT OF COLUMBIA • MARYLAND
www.SETTY.com
PROJECT NUMBER: SA2017-F-8-F-F

11720 Beltsville Drive
Suite 600
Calverton, MD 20705
Tel: 301.595.1000
www.grimmandparker.com
GRIMM AND PARKER
GP# 21553

FIRE PROTECTION DETAILS

HOLABIRD ACADEMY PK-8
CITY OF BALTIMORE, MARYLAND

DATE	DESCRIPTION

FP-7.1

03/13/2017
BID SET

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